torch.cumprod#

https://pytorch.org/docs/stable/generated/torch.cumprod.html#torch.cumprod

Description:

Returns the cumulative product of elements of input in the dimension dim. For example, if input is a vector of size N, the result will also be a vector of size N, with elements. input (Tensor) – the input tensor. dim (int) – the dimension to do the operation over dtype (torch.dtype, optional) – the desired data type of returned tensor. If specified, the input tensor is casted to dtype before the operation is performed. This is useful for preventing data type overflows. Default: None.

Function Signature

```
torch.cumprod(input, dim, *, dtype=None, out=None) → Tensor#
Parameters
Keyword Arguments
```

Parameters

Keyword

dtype (torch.dtype, optional) – the desired data type of returned tensor. If specified, the input tensor is casted to dtype before the operation is performed. This is useful for preventing data type overflows. Default: None. out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.randn(10)
>>> a
tensor([ 0.6001,  0.2069, -0.1919,  0.9792,  0.6727,  1.0062,
0.4126,
        -0.2129, -0.4206,  0.1968])
>>> torch.cumprod(a, dim=0)
tensor([ 0.6001,  0.1241, -0.0238, -0.0233, -0.0157, -0.0158,
-0.0065,
         0.0014, -0.0006, -0.0001])

>>> a[5] = 0.0
>>> torch.cumprod(a, dim=0)
tensor([ 0.6001,  0.1241, -0.0238, -0.0233, -0.0157, -0.0000,
-0.0000,
         0.0000, -0.0000, -0.0000])
```

Detailed Documentation

Documentation

Returns the cumulative product of elements of input in the dimension dim.

For example, if input is a vector of size N, the result will also be a vector of size N, with elements.

input (Tensor) – the input tensor.

dim (int) – the dimension to do the operation over

dtype (torch.dtype, optional) – the desired data type of returned tensor. If specified, the input tensor is casted to dtype before the operation is performed. This is useful for preventing data type overflows. Default: None.

out (Tensor, optional) – the output tensor.